

Verb–Adverb Order in Trøndersk Embedded Clauses: An Exploratory Sociolinguistic Study

Ådne SKJÆVELAND JØSSING

adne.jossing@student.uni-tuebingen.de

Conducting Sociolinguistic Research (WiSe 2025-26)

Universität Tübingen

Abstract

This study explores the distribution of verb–adverb vs. adverb–verb ordering in embedded clauses in spontaneous speech from nine speakers of Trøndersk Norwegian. It examines gender, age, structural distance, and clause type as potential conditioning factors. Data were collected through semi-structured interviews and annotated manually. None of the three primary hypotheses were supported: women produced non-canonical ordering at a higher rate than men, no meaningful age effect was found, and the distance hypothesis showed no clear result. The clearest finding is a strong distributional pattern across clause types, with complement clauses introduced by *at* showing the highest rates of non-canonical ordering. Most other clause types show near-categorical canonical ordering.

1 Introduction

Word order in human languages reflects the interaction of syntax, semantics, and discourse. In Norwegian, as well as the other Scandinavian languages Swedish and Danish, main clause word order places the finite verb in second position (V2). In dependent clauses however, the finite verb canonically follows adverbials. This pattern is generally adhered to in standard writing, but non-canonical ordering in dependent clauses, with verbs preceding adverbials, occurs frequently in spontaneous speech.

Earlier research has found that verb placement in embedded clauses is affected by both structural configuration and semantic properties of the matrix predicate. This was initially documented in English, later in Scandinavian (Hooper and Thompson, 1973; Vikner,

1994). More recently, Djärv et al., 2017 identifies assertion and factivity as the primary semantic environments conditioning embedded verb–adverb in Scandinavian, and corpus studies such as Ringstad, 2019 demonstrate that the variation is distributed systematically. Despite this, inter-speaker variation in embedded verb placement has received little quantitative study. The present study investigates distributional patterns of adverb–verb vs. verb–adverb ordering in embedded clauses in Trøndersk Norwegian, examining both social conditioning factors (gender, age) and a structural factor (distance between subordinator and verb), as well as giving an overview of the distribution across embedded clause types.

1.1 Hypotheses

The following study tests three primary hypotheses. The first and second hypotheses address social variation, and the third hypothesis explores linguistic conditioning:

- H1. Women more strongly prefer the canonical adverb–verb ordering in embedded clauses than men.
- H2. Older speakers more strongly prefer the canonical adverb–verb ordering in embedded clauses than younger speakers.
- H3. Non-canonical verb–adverb ordering in embedded clauses is more likely when the embedded verb is further from the subordinating element.

The first hypothesis aligns with literature showing that women often lead the use of standard variants in variable phenomena. The second hypothesis follows from the general expectation that older speakers tend to favour more conservative, established grammatical norms. The third hypothesis builds on the assumption that increased linear distance between the subordinating element and the embedded verb may correlate with a weaker link between them, thereby increasing the likelihood of non-canonical ordering.

2 Background and Related Work

The distribution and interpretation of embedded clauses have been subject to extensive research, both in English and in the Scandinavian languages. A central contribution that has inspired most subsequent work is Hooper and Thompson, 1973, who find that the syntactic behaviour of embedded clauses is conditioned by semantic properties of their matrix predicates. Only predicates that assert the truth of their complements, such as *say*, *think*, and *believe*, can reliably allow embedded clauses to contain syntactic phenomena

otherwise limited to main clauses, while factive predicates such as *regret* and *be surprised*—only presupposing the truth of their complements—do not allow this. Variables investigated in their research on English include topicalization and negative preposing.

In Scandinavian languages the distinction between main and embedded clauses is morphosyntactically clearer than in English when sentential adverbs and negation are involved: main clauses display verb–adverb order, while embedded clauses canonically display adverb–verb order (Vikner, 1994). However, in spontaneous speech certain embedded clauses are instead verb–adverb, like in main clauses. Building on Hooper and Thompson, 1973 and the concept of the Main Point of Utterance (MPU)—describing the proposition a speaker primarily commits to in an utterance (Simons, 2007)—Djäv et al., 2017 argue that embedded verb–adverb order in Scandinavian is licensed when the embedded clause constitutes the MPU. This means that only complement clauses under assertive predicates allow verb–adverb order, while factive or presupposed complements do not allow it. Examples of permitted and non-permitted embedded main clause-like ordering are given below:

(1) *Asserted complement*

- a. Han sa [at han **ikkje** tok eksamen]
he said that he not took exam
 ‘He said that he didn’t take the exam.’
- b. Han sa [at han tok **ikkje** eksamen]
he said that he not took exam
 ‘He said that he didn’t take the exam.’

(2) *Factive Predicate*

- a. Han angra på [at han **ikkje** tok eksamen]
he regretted on that he not took exam
 ‘He regretted that he didn’t take the exam.’
- b. *Han angra på [at han tok **ikkje** eksamen]
he regretted on that he not took exam
 ‘He regretted that he didn’t take the exam.’

One important factor to note from this framework lies in the distribution of embedded verb–adverb ordering across clause types. The assertive/factive distinction is primarily defined for complement clauses introduced by *at*, where the embedded content can either be asserted or presupposed depending on the matrix predicate. Relative clauses introduced by *som*, on the other hand, identify a referent rather than assert a proposition,

and are therefore outside the assertive/factive paradigm. Similarly, indirect questions are non-assertive by definition, and adverbial clauses introduce circumstances rather than propositions to be asserted or presupposed. The framework therefore predicts that embedded verb–adverb ordering should be concentrated in *at*-complement clauses, and is rare or absent in other embedded clause types. The present study examines this prediction as part of a broader investigation of the distribution of embedded clause ordering across subordinator types.

Ringstad, 2019 provides quantitative perspectives, examining the distribution of embedded verb–negation order in Norwegian using corpus data, and concludes that non-canonical verb placement is not random variation, but rather reflects a systematic interaction between negation placement and clause type.

While the conditioning of embedded verb–adverb has received considerable attention, comparatively little work has been done on inter-speaker variation. Bentzen, 2007 is a notable exception, investigating verb movement in embedded clauses across three Norwegian varieties, and finds systematic dialect-based differences. Eastern Norwegian requires canonical adverb–verb order in all embedded clauses, Tromsø Northern Norwegian allows finite auxiliaries to precede adverbs, and Regional Northern Norwegian allows all finite verbs to precede adverbs freely. Beyond this dialect-based stratification, systematic investigation of other inter-speaker factors, such as age and gender, remains absent from the literature.

The study of such factors in sociolinguistics has a well-established theoretical foundation in the work of Labov (1972, 1990). Labov identifies a consistent pattern in stable sociolinguistic variables, where women tend to favour prestige or standard variants over non-standard variants. However, in cases of ongoing change from below, where a non-standard variant is spreading through the community without overt social marking, women often lead this change rather than resist it. This distinction between stable variation and change in progress is relevant to the present study, since it is not established whether verb–adverb order in embedded clauses represents a stable variable, or a feature currently increasing or decreasing in use among speakers of Trøndersk.

The present study addresses the gap in research on inter-speaker variation. Building on the semantic conditioning established by Djärv et al., 2017 and the quantitative baseline provided by Ringstad, 2019, it investigates the distribution of embedded verb–adverb order in nine speakers of Trøndersk, with particular attention to gender, age, and the distribution across embedded clause types. The aim is to determine whether social and structural factors beyond clause type and matrix predicate semantics can account for ob-

served variation, and to examine whether the distributional patterns across clause types are consistent with the predictions of Djärv et al., 2017, integrating sociolinguistic dimensions into the phenomenon to a larger extent than what has been done before.

3 Data and Methodology

3.1 Participants

Data were collected from nine speakers of Trøndersk, a dialect of Norwegian spoken in the Trøndelag region of central Norway. All participants are lifelong residents of the region with no extended periods of residence outside the dialect area. Participants can be divided into two gender groups and three age groups: young (18–23), middle (54–56), and old (72–82). Information on all participants, including age, gender, education level, and interview duration is given in Table 1.

ID	Gender	Age	Age group	Occupation	Education	Duration
P1	Female	18	Young	Student	Upper secondary	44 min.
P2	Male	19	Young	Student	Upper secondary	45 min.
P3	Male	23	Young	Student	Upper secondary	45 min.
P4	Female	54	Middle	Administrator	Master’s	48 min.
P5	Male	55	Middle	Professor	Doctorate	49 min.
P6	Female	56	Middle	Teacher	Bachelor’s	47 min.
P7	Male	72	Old	Retired	Bachelor’s	44 min.
P8	Female	80	Old	Retired	Upper secondary	44 min.
P9	Male	82	Old	Retired	Upper secondary	45 min.

Table 1: Speaker profiles

The sample size of nine participants is small. It is, however, sufficient for an exploratory study of this kind, where the goal is to recognize patterns of syntactic variation across demographic groups, rather than to establish statistically generalizable results (Milroy, 2003).

3.2 Interview Design

Data were collected through semi-structured interviews of approximately 45 minutes each. The first 15 minutes of each interview were not used in the analysis, in order to minimize the effect of the observer’s paradox: participants tend to monitor their speech more carefully in the beginning of a recorded interview, and discarding this portion allows the analysis to focus on more natural, unmonitored speech (Labov, 1990). The

interviews were conducted in Norwegian and consisted primarily of open-ended questions with short follow-ups to encourage extended spontaneous speech. Interruptions were avoided and the interviewer’s own contributions were kept minimal, to maximize the proportion of natural speech from the participant.

The interviews were structured around two main topics: work and leisure. This division follows the topic-based approach to formality stratification from Labov, 1981, under which discussion of work tends to elicit more careful and monitored speech, while discussion of leisure activities tends to elicit more relaxed, vernacular speech. This allows for comparison not only across speaker groups, but also across degrees of formality within the same speaker. In addition to the two main topics, participants were asked to recount a recent personal experience shared with another person, as narrative tasks of this kind tend to reduce conscious attention to linguistic form.

3.3 Elicitation of Embedded Clauses

Relative clauses account for only a small portion of speech, especially those including sentential adverbs or negation. In data from Ringstad, 2019 only 6% of utterances contain an embedded clause. Those with negation are rarer, constituting only 0.45% of all utterances. Therefore, several strategies were employed to increase the yield of embedded clauses in the data. Opinion elicitation questions such as *Kva trur du om ...?* ‘What do you think about ...?’ and evaluative follow-ups such as *Angra du på det?* ‘Did you regret that?’ were used to prompt embedded clauses under both assertive and factive predicates, the two contexts most directly relevant to the theoretical framework of Djärv et al., 2017. Relative clauses were elicited by asking participants to describe people, places, and objects.

3.4 Data Processing and Annotation

Initially, automatic annotation was attempted using *Tale til Tekst*, a service provided by NTNU¹ for automatic transcribing of audio and video material, but this was found to be insufficiently accurate to rely on, despite being trained on Norwegian dialectal data, including Trøndersk. Annotation of all sentences containing embedded clauses with sentential adverbs and negation was therefore carried out manually. Sentences in which the only adverb present was an epistemic or evidential adverb, such as *jo*, *nok*, and *vel*, were excluded, as the distribution of these elements in embedded clauses is governed by separate constraints outside the scope of this study. For each annotated sentence,

¹Norwegian University of Science and Technology

the following variables were coded: embedding verb, embedding element, finite verb, sentential adverbs, and the relative order of the latter two. A sample of the annotated data is shown in Table 2.

ID	Time	Order	Annotation
P2	17:01	V-Adv	Eg e(veit) r(at) eg a(kanskje) v(treng) det.
P4	23:16	Adv-V	Eg e(skjønnte) r(at) det v(var) a(egentlig) ganske artig.

Table 2: Sample annotation lines. Annotation tags: e() = embedding verb, r() = embedding element, v() = embedded finite verb, a() = sentential adverb.

Due to the interviews containing dialectal speech with no standardized orthography and minor inter-speaker morphological differences, all examples drawn from the data presented in this paper are represented in Nynorsk, a written standard of Norwegian.

3.5 Statistical Analysis

Group comparisons of verb–adverb rates were conducted using two tests. Fisher’s exact test was used to compare raw pooled counts between groups, and Mann-Whitney U tests were used for speaker-normalized comparisons. Logistic regression models were also fitted to observe the effect of structural distance on ordering, with gender and age group included as covariates. All statistical analyses were conducted in Python with the SciPy library (Virtanen et al., 2020).

4 Results and Discussion

A total of 212 sentences containing embedded clauses with sentential adverbs were identified across the nine interviews, after excluding sentences with epistemic and evidential particles. Of these 212, 156 (73.6%) displayed canonical adverb–verb order, while 56 (26.4%) displayed non-canonical verb–adverb order. This baseline rate of non-canonical ordering is broadly consistent with the corpus findings of Ringstad, 2019, who reports approximately 33% verb–negation order in permissive contexts. Considering the sentences from the present study also include impermissive contexts, this lends initial support to the validity of the data collection and annotation procedure.

Individual speaker rates varied considerably, ranging from 8.6% verb–adverb (P5, Male, Middle) to 45.8% (P4, Female, Middle), as summarized in Table 3. This degree of inter-speaker variation suggests that there is a great degree of individual variation, and that group-level patterns should be interpreted with caution given the small sample size.

ID	Gender	Age group	V-Adv	Total	% V-Adv
P1	Female	Young	6	24	25.0
P2	Male	Young	4	23	17.4
P3	Male	Young	6	21	28.6
P4	Female	Middle	11	24	45.8
P5	Male	Middle	3	35	8.6
P6	Female	Middle	7	25	28.0
P7	Male	Old	5	21	23.8
P8	Female	Old	11	25	44.0
P9	Male	Old	3	14	21.4
Total			56	212	26.4

Table 3: V-Adv rates by speaker

Because sentence counts differ substantially across speakers, ranging from 14 (P9) to 35 (P5), raw pooled counts risk over-representing high-output speakers. This effect would be particularly strong since P5 is an outlier not only in the count of qualifying phrases, but also his low number of non-canonical verb–adverb orderings. Due to this, all group comparisons below report both raw pooled rates and speaker-normalized mean rates, where each speaker contributes equally regardless of total output.

4.1 H1: Gender and Verb–Adverb Order

The first hypothesis predicted that women would more strongly prefer canonical adverb–verb ordering than men. The data do not support this prediction. Female speakers produced verb–adverb order at a substantially higher rate than male speakers, both in raw pooled counts (35.7% vs. 18.4%), and in speaker-normalized means (35.7% vs. 20.0%). Fisher’s exact test on pooled counts yielded $p = 0.005$ with an odds ratio of 2.46, meaning that the female speakers in this sample were approximately two and a half times as likely to produce verb–adverb order as male speakers. The Mann-Whitney U test on per-speaker rates yielded $p = 0.064$. This result falls short of conventional significance thresholds, which is unsurprising given that only four female speakers contribute to the comparison. The test has very low statistical power at this sample size. The directional consistency between the two analyses nonetheless suggests the gender difference is not an artefact of pooling.

The original hypothesis of women using non-canonical ordering drew on the observation from Labov, 1990 that women tend to favour prestige variants in stable sociolinguistic variables. However, Labov also identifies a complementary pattern: in changes from below, where a non-standard variant is spreading through the community, women

frequently lead the change. If verb–adverb order in embedded clauses is not a receding stigmatized form, but rather a spreading vernacular feature, the data would be consistent with this second pattern. Alternatively, due to the low sample size, the finding may also simply reflect individual speaker differences not reducible to gender, as two of the four female speakers (P4 and P8) account for a disproportionate share of female verb–adverb tokens, with rates of 45.8% and 44.0% respectively, while the other two female speakers pattern much more similarly to the male group. With such a low number of speakers it is not possible to distinguish a systematic gender effect from high individual variation.

4.2 H2: Age and Verb–Adverb Order

The second hypothesis predicted that older speakers would more strongly prefer canonical adverb–verb ordering than younger speakers. The data do not support this prediction either. Speaker-normalized mean verb–adverb rates increase slightly across age groups: 23.7% in the young group, 27.5% in the middle group, and 29.7% in the old group. While the direction of this trend is opposite of the hypothesis, the difference between groups is small, and the within-group variance is very high, meaning that the age groupings cannot be taken to reflect any real tendencies.

4.3 H3: Distance and Verb–Adverb Order

The third hypothesis predicted that verb–adverb ordering would be more likely the further the verb–adverb pair was from the subordinating element. Before this can be measured, there is one important consideration to take into account. In canonical adverb–verb embedded clauses the adverb sits between the subordinating element and the verb, inflating the distance measure for canonical sentences relative to non-canonical ones. The distance measure was therefore computed excluding adverb tokens, counting only non-adverbial intervening material between the subordinating element and the finite verb. This allows the distance to be measured independently of the variable element whose position is investigated.

Mean distance is 1.44 for canonical adverb-verb sentences and 1.10 for non-canonical verb–adverb sentences, a difference not large enough to be meaningfully interpreted. The Mann-Whitney U test is non significant ($p = 0.849$), and the point-biserial correlation between distance and order is $r = -0.098$ ($p = 0.232$). Logistic regression confirms this: corrected distance is not a significant predictor of word order ($\beta = -0.211$, $p = 0.245$, pseudo- $R^2 = 0.010$). The effect does not reach a significance in any of the regression models, and the corrected distance coefficient remains stable and non-significant whether or

not gender and age group are included as covariates. H3 can therefore not be conclusively supported.

The distribution by corrected distance shows no coherent ordering tendency. At distance 0, 20.0% of sentences display verb–adverb order, at distance 1, 29.4%, at distance 2, 57.1%. This last observation should not be over-interpreted, as it is based only on seven sentences. At distances of 3 and above all instances are from P5, and all have adverb–verb order. P5 is an outlier in this statistic as well, in having embedded clauses with considerably longer distances between the embedding element and the embedded verb than other participants. Therefore, no clear monotonic relationship between distance and order probability can be observed.

Something to be noted is that gender emerges as a potential predictor in the regression models. In the model that measures distance and gender, the female coefficient approaches significance ($\beta = 0.736, p = 0.058$). In the model which additionally includes age group, gender reaches conventional significance ($\beta = 0.809, p = 0.043$) and is the only significant predictor in the model. This is consistent with the findings for H1, and seems to suggest that gender is a modest predictor of the variance in word order.

The lack of clear results in the effect of distance in predicting verb–adverb ordering should not be interpreted as evidence that structural factors play no role in embedded clause ordering, but rather reflects the limits of what can be measured from naturalistic interview data with a small corpus. Most sentences in the dataset cluster at distances of 0–1, leaving little variance for a distance predictor to explain. A study giving participants acceptability judgement tasks on structural distance with controlled sequence materials, would give cleaner conditions to better test for the effect of distance on embedded clause verb–adverb ordering, similar to the one conducted in Djärv et al., 2017.

4.4 Subordinator Type and Clause Category

Table 4 shows verb–adverb rates broken down by subordinating element. The results pattern clearly along clause type lines and are broadly consistent with the predictions of Djärv et al., 2017, with a few notable exceptions.

Complement clauses introduced by *[at]* (underlying *at* absent from surface structure) display the highest rate of non-canonical ordering among subordinators, at 43.8%. Clauses introduced by *at* are predicted to license verb–adverb order under assertive predicates (Djärv et al., 2017). The data collected here cannot quantify the assertive/factive split directly, as individual assertiveness judgements for naturalistic speech data are not straight-

forward to assign. Additionally, the literature reaches no consensus on exactly which matrix predicates and contexts allow embedded verb–adverb order.

Subordinator	Adv–V	V–Adv	Total	% V–Adv
<i>[at]</i>	9	7	16	43.8
<i>fordi</i>	13	9	20	40.9
<i>at</i>	55	32	87	36.8
<i>sjølv om</i>	4	2	6	33.3
<i>der</i>	4	2	6	33.3
<i>[som]</i>	9	2	11	18.2
<i>som</i>	33	2	35	5.7
<i>når</i>	12	0	12	0.0
<i>om</i>	4	0	4	0.0
<i>kva</i>	2	0	2	0.0

Table 4: The ten most frequently used subordinators ordered by frequency of V–Adv order. Zero complementizers have their underlying complementizer in square brackets.

Embedded clauses introduced by *som* have a much lower rate, at 5.7%, which is expected under Djärv’s framework: these simply identify a referent. There is therefore no discourse motivation for the embedded clause to behave like a main clause.

Temporal adverbials (*når*) and conditional clauses (*om*) similarly show 0% verb–adverb. Embedded question complementizers such as *kva* and *kor* pattern the same way. This is consistent with background literature, as such complementizers introduce circumstantial content, which would not constitute the MPU.

Complementizer Ellipsis

Sentences in which the complementizer is present in the underlying syntactic structure but absent from the surface realization form a notable pattern. These are annotated with *[at]* and *[som]*, with the underlyingly present complementizers put in square brackets. These display verb–adverb rates of 43.8% and 18.2%, higher than their overt counterparts at 36.8% and 5.7%, respectively. The subordinating element being absent in the surface realization makes the clause boundary less explicitly marked, which may reduce the strength of the syntactic cue that the embedded clause is a subordinate environment. This could make main clause-like ordering more accessible to the speaker, since the local syntactic context more closely resembles a main clause. This interpretation is consistent with proposals that overt complementizers serve as explicit cues to signal embedded clause structure (Jaeger, 2010).

Causal Clauses

One notable finding is the high verb–adverb rate under the causal subordinator *fordi* (40.9%). This is not covered by the Djärv framework, but one possible explanation is that causal clauses in spontaneous speech often carry strong informational weight: the clauses are often new, and the content that the speaker is primarily communicating. With this interpretation the main function of *fordi*-clauses is to introduce the MPU, and they therefore license main-clause-like syntax. Ringstad, 2019 finds a comparable distribution of *fordi*-clauses in her data, and rationalizes it similarly. It should also be noted that the gender breakdown for *fordi* is large: 80.0% verb–adverb for female speakers, and 0% among male speakers.

Gender and Age with Subordinator Type

Differences in gender and age cannot meaningfully predict the distribution of clause types used by speakers, as relative frequencies of use for different clause types are comparable across gender and age. A large portion of subordinators only occur once or twice in the dataset, leaving no room for meaningful interpretation.

4.5 Formality and Reported Speech

The interview design distinguished three discourse contexts: discussion of work, discussion of leisure, and narrative accounts of recent personal experiences with other people. No clear difference in verb–adverb rates could be observed between the work and leisure conditions, and the boundary between the two was in practice difficult to maintain consistently across the interviews, as conversations moved naturally between topics. No systematic analysis of the formality split is therefore reported here.

One qualitative observation from the narrative condition is worth noting, however. When participants recounted experiences involving interactions with other people, embedded clauses functioning as indirect reported speech—constructions where the content of another person’s utterance is reported as an embedded complement rather than quoted directly, as in *og da sa han at han hadde ikkje gjort det* ‘and then he said that he hadn’t done it’—seemed to contain non-canonical verb–adverb order more frequently than other embedded clauses. This might be caused by the main communicative content of the utterance here being carried by the embedded clause, making it a strong candidate for MPU status. The boundary between indirect reported speech and direct speech may also be less sharply delimited in spontaneous spoken language than in writing. This could be pulling indirect speech complements toward main-clause-like syntactic structure.

The participant P5 is an outlier across nearly all measures, in having the lowest verb–adverb rate in the sample, the longest mean distance between subordinator and verb, and the highest count of sentences containing the relevant variable. P5 has a doctoral degree and works as a university professor, and his professional work with formal written Norwegian may contribute to a more consistently canonical embedded clause grammar, which is consistent with findings on the relationship between education and prestige variant use (Labov, 1972).

5 Conclusion

This study has investigated the distribution of verb–adverb vs. adverb–verb ordering in embedded clauses in spontaneous speech data from nine speakers of Trøndersk Norwegian, with a focus on gender, age, structural distance, and variation across embedded clause types. The overall rate of non-canonical verb–adverb ordering is largely consistent with distributions found in other studies, and suggests that the phenomenon is present in Trøndersk in largely the same way as in other Norwegian varieties.

Neither H1 nor H2 was supported in the predicted direction. Female speakers produced the verb–adverb order at a higher rate than male speakers, the opposite of what was predicted, and the effect was statistically significant in pooled counts but could not be deemed significant in the speaker-normalized analysis. Age showed no decisively interpretable effect in either direction, with high variance within each age group. H3 was also not supported: no significant relationship between structural distance and ordering was found.

The clearest finding is the distributional pattern across clause types. Complement clauses introduced by *at* have the highest rate of verb–adverb ordering, followed by relative clauses introduced by *som*, while temporal adverbials and embedded questions show canonical ordering only. This is consistent with the framework from Djärv et al., 2017. The higher verb–adverb rates under zero complementizer constructions compared to their overt counterparts are consistent with the proposal that overt subordinating elements serve as explicit cues to embedded clause structure, and therefore reduce the chance of main-clause-like ordering (Jaeger, 2010). Taken together, the results suggest that clause type is a more reliable predictor of embedded verb placement than the linguistic and social constraints investigated, at least at this sample size.

5.1 Limitations

The main limitation of the study is the sample size. A sample of nine participants is insufficient to draw clear conclusions about gender or age effects. Even though the observed gender difference is consistent across analyses, this could simply reflect individual grammatical preferences of two high-rate speakers (P4 and P8), who happen to be female. This cannot be determined with the low amount of data used in this study.

The study is additionally limited in that assertive/factive conditioning could not be quantified directly, since assigning assertiveness judgements to individual utterances is not straightforward, and also lacks consensus criteria in the literature. The clause type analysis gives insight into this conditioning, but a direct test of theoretical predictions against spoken data remains to be done.

5.2 Future Directions

Several directions for future work could follow from this study. First, a larger study would provide the statistical certainty needed to determine if the gender effect is genuine, and to assert the lack of a difference between different age groups. Second, a follow-up with acceptability judgement tasks with controlled sentence materials could allow the distance hypothesis to be tested under cleaner conditions, where structural distance can be changed independently of other factors. Third, a study combining systematic annotation of matrix predicate assertiveness with a consistent treatment of indirect reported speech as a separate variable would allow the predictions of Djärv et al., 2017 to be tested directly against naturalistic spoken data, and contribute to a more complete understanding of verb–adverb order in Trøndersk embedded clauses.

Acknowledgements

I wish to thank all nine participants for their time and willingness to share their experiences and thoughts during the interviews. Their contributions made this study possible. I also wish to thank Karen V. Beaman for organizing and teaching the course *Conducting Sociolinguistic Research*, which made me think about and explore aspects of language I had not previously considered.

References

- Bentzen, Kristine (2007). "The degree of verb movement in embedded clauses in three varieties of Norwegian". In: *Nordlyd* 34.1.
- Djävrv, Kajsa, Caroline Heycock, Hannah Rohde, Laura Bailey, and Michelle Sheehan (2017). "Assertion and factivity: Towards explaining restrictions on embedded V2 in Scandinavian". In: *Order and structure in syntax I*, p. 3.
- Hooper, Joan B and Sandra A Thompson (1973). "On the applicability of root transformations". In: *Linguistic inquiry* 4.4, pp. 465–497.
- Jaeger, T Florian (2010). "Redundancy and reduction: Speakers manage syntactic information density". In: *Cognitive psychology* 61.1, pp. 23–62.
- Labov, William (1972). "Some principles of linguistic methodology". In: *Language in society* 1.1, pp. 97–120.
- Labov, William (1981). "Field Methods of the Project on Linguistic Change and Variation. Sociolinguistic Working Paper Number 81." In.
- Labov, William (1990). "The intersection of sex and social class in the course of linguistic change". In: *Language variation and change* 2.2, pp. 205–254.
- Milroy, James (2003). "On the role of the speaker in language change". In: *Motives for language change*, pp. 143–157.
- Ringstad, Tina Louise (2019). "Distribution and function of embedded V–Neg in Norwegian: A corpus study". In: *Nordic Journal of Linguistics* 42.3, pp. 329–363.
- Simons, Mandy (2007). "Observations on embedding verbs, evidentiality, and presupposition". In: *Lingua* 117.6, pp. 1034–1056.
- Tagliamonte, Sali A (2006). *Analysing sociolinguistic variation*. Cambridge University Press.
- Vikner, Sten (1994). "Scandinavian embedded clauses". In: *Verb movement*, p. 117.
- Virtanen, Pauli et al. (2020). "SciPy 1.0: fundamental algorithms for scientific computing in Python". In: *Nature methods* 17.3, pp. 261–272.